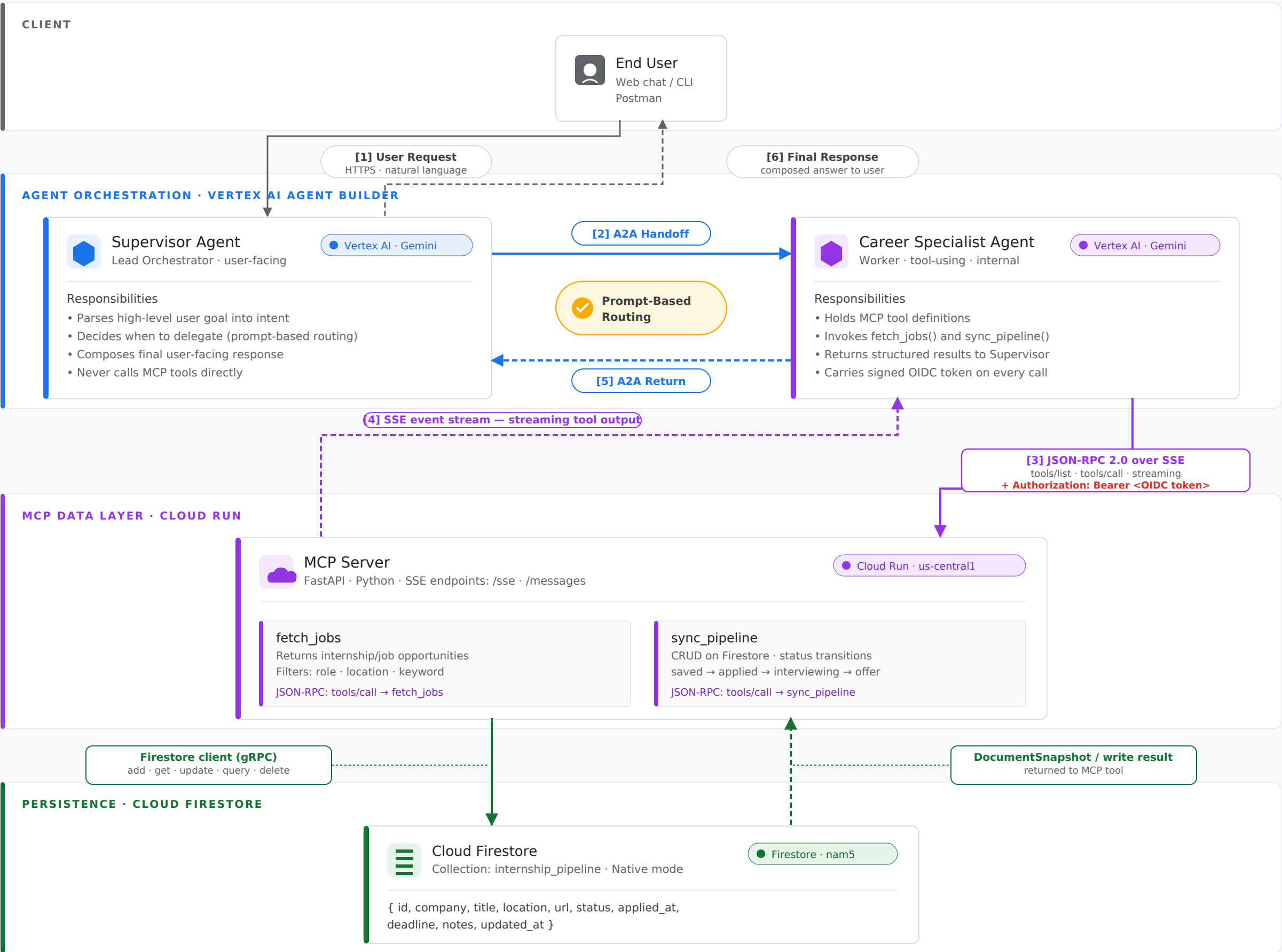


Agentic Career Coach (ACC) — System Architecture

Supervisor-Specialist Multi-Agent System on Google Cloud Platform

INFO 520 · Capstone Project
Roshan & Michael · VCU · Spring 2026
Submission 1 — Architecture



REFERENCE

CONNECTIONS

- Black arrow: User ↔ Supervisor (HTTPS / TLS)
- Blue arrow: A2A Handoff (Supervisor ↔ Specialist)
- Purple arrow: JSON-RPC 2.0 over SSE (Specialist ↔ MCP Server)
- Green arrow: Firestore gRPC (MCP Server ↔ Firestore)
- Dashed line: Response path (return trip on every hop)

REQUEST FLOW

- 1 User → Supervisor
- 2 A2A handoff to Specialist
- 3 MCP tool call (SSE)
- 4 SSE stream returns result
- 5 A2A return to Supervisor
- 6 Supervisor replies to user

PROTOCOLS

- Model Context Protocol
- JSON-RPC 2.0
- Server-Sent Events
- OIDC (RFC 7519)
- gRPC (Firestore)

GCP SERVICES

- Vertex AI Agent Builder
- Cloud Run
- Cloud Firestore
- Cloud IAM + OIDC
- Cloud Logging + Trace

DOCUMENT

INFO 520 · Data Communications
Capstone Submission 1
Roshan & Michael · VCU · Apr 2026

Architecture diagram deliverable
for multi-agent ACC system.

ZERO-TRUST SECURITY · IAM + OIDC (EXTRA CREDIT +10)

Access: Cloud Run requires authentication · no public ingress

Identity: Signed OIDC ID token in Authorization: Bearer header

Result: Unauthenticated calls → HTTP 403 · per-request verification